```
import matplotlib.pyplot as plt
import numpy as np
from sklearn.datasets import load_iris
from sklearn.metrics import accuracy_score, classification_report
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC

iris = load_iris()


X = iris.data[:, :2]


y = (iris.target == 0).astype(int)


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)


scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)


model = SVC(kernel="linear", C=1.0)
model.fit(X_train, y_train)


y_pred = model.predict(X_test)

print("--- MODEL PERFORMANCE ---")
print("Accuracy:", round(accuracy_score(y_test, y_pred) * 100, 2),
"%")
print("Support Vector Count per Class [Non-Setosa, Setosa]:",
model.n_support_)
print("\nClassification Report:")
print(
    classification_report(
        y_test, y_pred, target_names=["Non-Setosa", "Setosa"],
zero_division=0
    )
)

x_min, x_max = X_train[:, 0].min() - 0.5, X_train[:, 0].max() + 0.5
y_min, y_max = X_train[:, 1].min() - 0.5, X_train[:, 1].max() + 0.5

xx, yy = np.meshgrid(
    np.linspace(x_min, x_max, 300),
```

```
    np.linspace(y_min, y_max, 300)
)


Z = model.decision_function(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)


plt.figure(figsize=(10, 6))


plt.contourf(
    xx, yy, Z, levels=[-100, 0, 100], colors=["lightblue",
"lightcoral"], alpha=0.3
)


plt.contour(
    xx,
    yy,
    Z,
    levels=[-1, 0, 1],
    colors=["dimgray", "black", "dimgray"],
    linestyles=["--", "-", "--"],
    linewidths=[1.5, 2.0, 1.5],
)

plt.scatter(
    X_train[y_train == 0, 0],
    X_train[y_train == 0, 1],
    label="Train: Non-Setosa",
    marker="o",
    c="blue",
    alpha=0.7,
)
plt.scatter(
    X_train[y_train == 1, 0],
    X_train[y_train == 1, 1],
    label="Train: Setosa",
    marker="^",
    c="red",
    alpha=0.7,
)


plt.scatter(
    X_test[:, 0],
    X_test[:, 1],
    c=y_test,
    cmap=plt.cm.coolwarm,
```

```
    marker="s",
    edgecolors="k",
    s=70,
    label="Test Samples",
)


plt.scatter(
    model.support_vectors_[:, 0],
    model.support_vectors_[:, 1],
    s=160,
    facecolors="none",
    edgecolors="black",
    linewidths=1.8,
    label="Support Vectors",
)

plt.xlabel("Standardized Sepal Length")
plt.ylabel("Standardized Sepal Width")
plt.title("Linear SVM: Decision Boundary, Margins & Test Evaluation")
plt.legend(loc="upper left")
plt.grid(True, alpha=0.3)
plt.tight_layout()

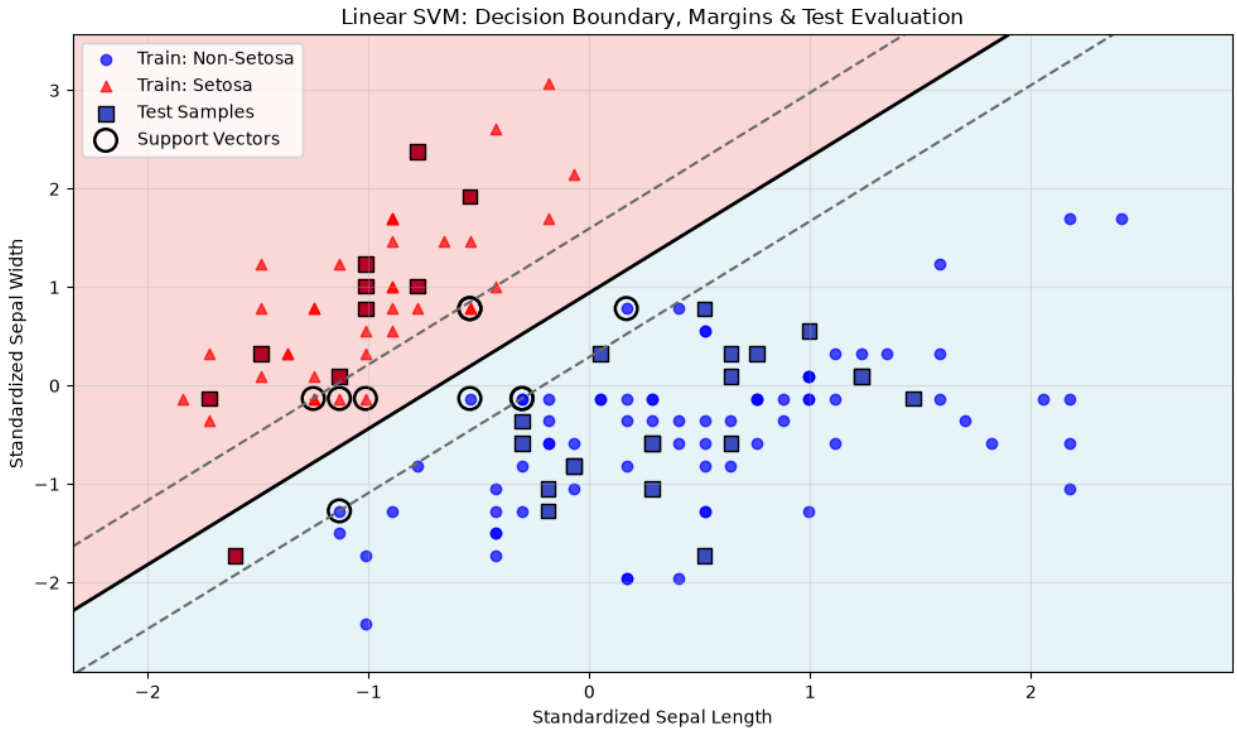

plt.savefig("svm_iris_graph.svg", format="svg", bbox_inches="tight")
plt.show()
```

```
--- MODEL PERFORMANCE ---
Accuracy: 96.67 %
Support Vector Count per Class [Non-Setosa, Setosa]: [5 5]

Classification Report:
              precision    recall  f1-score   support

  Non-Setosa       0.95      1.00      0.98        20
      Setosa       1.00      0.90      0.95        10

    accuracy                           0.97        30
   macro avg       0.98      0.95      0.96        30
weighted avg       0.97      0.97      0.97        30
```


Linear SVM: Decision Boundary, Margins & Test Evaluation
Train: Non-Setosa
Train: Setosa
Test Samples
Support Vectors
Standardized Sepal Width
Standardized Sepal Length
3
2
1
0
−1
−2
−2
−1
0
1
2